$[a]_I$ invertible in R/I iff $aR + I = R$

R/I field $\iff$ I maximal

$[a]_n$ invertible in $\mathbb{Z}/n\mathbb{Z}$ $\iff$ $\gcd(a,n)=1$ $\quad (n \in \mathbb{Z}_{\geq 2})$

$\mathbb{Z}/n\mathbb{Z}$ field $\iff$ n prime number

Summary of what we've had yesterday. Ad hoc examples.

$[f]_g$ invertible in $k[x]/g(x)k[x]$ $\iff$ $\gcd(f,g)=1$ $\quad (g \in k[x],\ \deg g \geq 1)$

$k[x]/g(x)k[x]$ field $\iff$ g irreducible

R

R/I

$I = g_1 R + \dots + g_s R$

$\mathbb{Z}$ is a PID (principal ideal domain)

$I = 6\mathbb{Z} + 4\mathbb{Z} = 2\mathbb{Z}$

invert $[5]_{12}$ in $\mathbb{Z}/12\mathbb{Z}$ (if invertible)

corresponds to $5\mathbb{Z} + 12\mathbb{Z} = \gcd(5,12)\mathbb{Z} = \mathbb{Z}$

$\Rightarrow$ yes, invertible

$5 \cdot 5 + 12 \cdot 2 = 1$

EEA:

$\begin{cases} a = 5 \\ n = 12 \end{cases}$

$-2a + n = 2$

$4a - 2n = 4$

$5a - 2n = 1$

$\Downarrow$

$[5]_{12}^{-1} = [5]_{12}$

1

$[8]_{12}$ invertible in $\mathbb{Z}/12\mathbb{Z}$?

$8\mathbb{Z} + 12\mathbb{Z} = \gcd(8,12)\mathbb{Z} = 4\mathbb{Z} \neq \mathbb{Z}$

$\Rightarrow$ $[8]_{12}$ non-invertible in $\mathbb{Z}/12\mathbb{Z}$

$[7]_{12}$ invertible in $\mathbb{Z}/12\mathbb{Z}$? $\rightsquigarrow$

$7\mathbb{Z} + 12\mathbb{Z} = \gcd(7,12)\mathbb{Z} = \mathbb{Z}$

$7 \cdot u + 12 \cdot v = 1$

$\downarrow$ EEA

$7 \cdot 7 + 12 \cdot (-4) = 1$

$[7]_{12}^{-1} = [7]_{12}$

$[11]_{12}^{-1} = [11]_{12}$

$12 - 11 = 1$

$12 \cdot 1 + 11 \cdot (-1) = 1$

$11 \cdot (-1) \equiv 1 \mod 12$

$[11]_{12} [-1]_{12} = 1$ in $\mathbb{Z}/12\mathbb{Z}$

$[11]_{12}$

$(\mathbb{Z}/12\mathbb{Z})^\times = \{[1]_{12}, [5]_{12}, [7]_{12}, [11]_{12}\}$ $\Longleftarrow$

By coincidence, in this ring one has

$x^2 = 1$ for every $x \in (\mathbb{Z}/12)^\times$. $\Longleftarrow$

1.8.19 Calculating the totient function from the prime factorization.

Remember: $\varphi(n) = \left|(\mathbb{Z}/n\mathbb{Z})^\times\right|$, $n \in \mathbb{Z}_{\geq 2}$

If we assume knowledge of the prime factorization of n, how can we compute $\varphi(n)$?

For example, $n = 3^2 \cdot 5^3 \cdot 7$. We know $\mathbb{Z}/n\mathbb{Z} \underset{\text{CRT}}{\overset{\sim}{=}} \mathbb{Z}/3^2\mathbb{Z} \times \mathbb{Z}/5^3\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z} \Rightarrow$

$$\varphi(n) = \left|(\mathbb{Z}/n\mathbb{Z})^\times\right| = \left|(\mathbb{Z}/3^2\mathbb{Z} \times \mathbb{Z}/5^3\mathbb{Z} \times \mathbb{Z}/7\mathbb{Z})^\times\right| = \left|(\mathbb{Z}/3^2\mathbb{Z})^\times \times (\mathbb{Z}/5^3\mathbb{Z})^\times \times (\mathbb{Z}/7\mathbb{Z})^\times\right| = \varphi(3^2)\cdot\varphi(5^3)\,\varphi(7).$$

$$\varphi(3^2) = \left|(\mathbb{Z}/3^2\mathbb{Z})^\times\right| = 3^2 - 3 = 3\cdot 2 = 6$$

$$\varphi(5^3) = \left|(\mathbb{Z}/5^3\mathbb{Z})^\times\right| = 5^3 - 5^2 = 5^2\cdot 4 = 100$$

$$\varphi(7) = \left|(\mathbb{Z}/7\mathbb{Z})^\times\right| = 7 - 1 = 6$$

$$\Rightarrow \varphi(n) = 3600.$$

0	1	2	3	4	5	6	7	8
x	v	v	x	v	v	x	v	v

0	1	2	3	4	5	6	7	8	9	10
x	v	v	v	v	x	v	v	v	v	x

Generally, when $n = p_1^{m_1} \cdots\cdots p_s^{m_s}$ is the prime factorization of n with primes $p_1, \dots, p_s$ where $p_i \neq p_j$ for $i \neq j$ and $m_1, \dots, m_s \in \mathbb{N}$, we have

$$\varphi(n) = \varphi(p_1^{m_1}) \cdot \ldots \cdot \varphi(p_s^{m_s}) = \prod_{i=1}^{s} p_i^{m_i - 1}(p_i - 1)$$

$$\varphi(p_i^{m_i}) = p_i^{m_i} - p_i^{m_i - 1} = p_i^{m_i - 1}(p_i - 1)$$

$$\underset{CRT}{\Longrightarrow} \boxed{z^{e+m(p-1)(q-1)} = z^e} \text{ for every } z \in R/_{I\cap J}.$$

So, the power functions $pow_e(z) = z^e$ satisfies $pow_{e+m(p-1)(q-1)} = pow_e$

So, in view this, the exponent can be associated with an element of $\mathbb{Z}/_{(p-1)(q-1)\mathbb{Z}}$.

1.9.2 Inverting the power function

Assume : $\gcd(e, (p-1)(q-1)) = 1$, then $e \cdot d = 1 + m \cdot (p-1) \cdot (q-1)$ for some $d, m \in \mathbb{N}$.

But then $z^{e \cdot d} = z^{1 + m \cdot (p-1)(q-1)} = z$ for all $z \in R/_{I\cap J}$. This means

$z^{ed} = pow_e(pow_d(z)) = pow_d(pow_e(z)) = z$. So the inverse of the power function $pow_e(z) = z^e$ is also a power function, this is the power function $pow_d(z)$.

1.9.3 Rivest - Shamir - Adleman cryptosystem